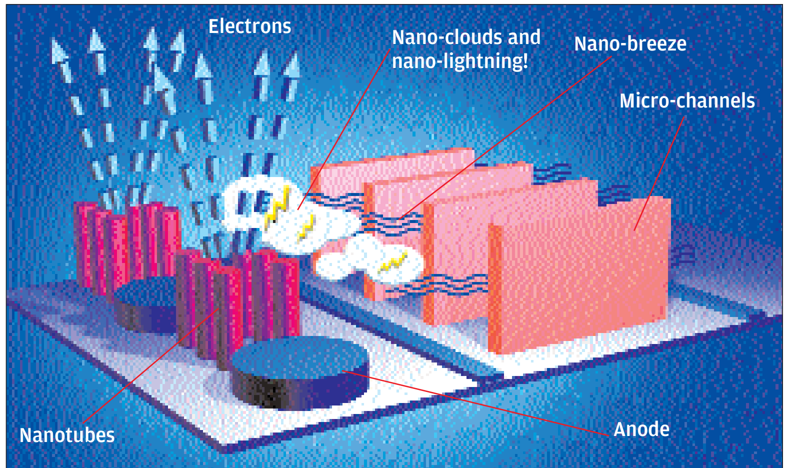

Cooling chips: the nanotubes release electrons that ionise the air and cause nano-lightning. As a result, nano-winds begin to flow in between the micro-channels, and cool the chip

Seagate Innovates... Again

Seagate is obsessed with hard disks. Which is quite in order, considering that's their business, but they have this habit of coming up with innovations ever so often. They're now dabbling in nanotubes, and are researching a rather complex process that can make hard disks denser.

We're saying "complex," but we can break it down into steps to see where the nanotubes come in.

First, we need denser hard disks. Because. Now, techniques such as perpendicular recording do make disks denser, but we need more density. Just because.

Now, if one just packs the bits on a hard disk platter too close together, there's a chance that a bit can flip its adjacent bit. In other words, it's demagnetisation at work, because of too high a density.

Now how does one remedy this? By using a recording material of high "anisotropy," meaning that it's harder to demagnetise, and so will reduce the chances of bits flipping each other. Problem: such material is also harder to *magnetise* as well.

This has an answer, too. Use a laser beam to *heat* the spot being recorded on, because when a spot is hot, it's easier to record on. This happens to be a better solution than using a stronger recording head.

Still with us? OK. When the heating is taking place, the lubricant film on top of the recording surface (yes, there is one) could evaporate or decompose. That, in turn, considerably reduces the life of the disk.

And here's where our hero—the nanotube—comes in. Spread nanotubes all over the surface of the platter—nanotubes filled with lubricant. The nanotubes slowly release the lubricant over the life of the disk, keeping the head spinning happily.

So that's how it is—they can hold lubricant as well, we've now learnt!

Aiding In Cooling...

...are the nanotubes of tomorrow. Cooling of chips, that is. Fujitsu and Intel are into this one, and it was reported more than a year ago.

What Intel is doing is simple. It just involves putting nanotubes into the thermal grease that makes up the layer between a microprocessor and its heatsink. A nanotube layer works *orders of magnitude* better than regular thermal paste, it's been claimed.

Why nanotubes? Because they conduct heat extremely well, and because they lend themselves to suspension in polymers and coatings. This second point is important: Intel will either design a polymer film containing billions of nanotubes, or try and find a way to deposit the nanotubes onto the (silicon) substrate.

Fujitsu's system is a heatsink made up of millions of nanotubes "grown" on a wafer substrate. The structure of the heatsink is such that it matches the pattern of the electrode bumps on the base of the chip to be cooled. Fujitsu is now working to improve the nanotube density around the bumps which, of course, will lead to even higher heat dissipation.

In independent research at Purdue University, "lightning" (!) produced by nanotubes could generate tiny air currents that can cool chips. This is air